

EduVision_DV

Higher Education Intelligence System for Institutional Performance & Academic Excellence

Unified Interactive Tableau Analytics Suite across 2,517 Global Universities & 105 Nations

Milestone 1-5 End-to-End Submission & Portfolio Showcase | 2026

1. Executive Summary & Core Objectives

- Project Statement: Transform disjointed higher education datasets into a unified, actionable intelligence system.
- Public Datasets Integrated:
 - * QS World University Rankings 2025 (1,503 institutions, 28 performance metrics)
 - * Times Higher Education (THE) World University Rankings 2024 (1,799 ranked institutions)
 - * World University Rankings (WUR) 2023 & World Bank EdStats country benchmarks
- Key Outcomes:
 - * Robust multi-source ETL with 95%+ completeness and zero duplicate entity corruption.
 - * 6 defensible higher education KPIs mathematically formulated without arbitrary approximations.
 - * Star-Schema Dimensional Data Model linking university dimensions, performance facts, and country metrics.
 - * 4 interconnected Tableau dashboards in a single production .twbx with full navigation & filter actions.

2. KPI Engineering & Mathematical Foundations

1. KPI 1 - Global Ranking Score:

Standardized overall institution score derived from QS Overall and cross-validated with THE Overall.

2. KPI 2 - Research Impact Score:

Citations per Faculty score directly measuring academic citation density (0-100 scale).

3. KPI 3 - Faculty-to-Student Ratio:

Formatted as an actual ratio string (1:X.X) backed by numeric Student-Staff ratio (Median: 1:17.3).

4. KPI 4 - International Student Percentage:

Explicit international cohort percentage (0-100%) preserving true educational diversity.

5. KPI 5 - Academic Reputation Score:

Peer-reviewed academic reputation index measuring institutional prestige and faculty esteem.

6. KPI 6 - Research Productivity Index (Composite Formulation):

Formula: $RPI = 0.50 * Research_Score_Norm + 0.30 * Citation_Score_Norm + 0.20 * Outlook_Score_Norm$

3. Interconnected Tableau Dashboard Suite

- Dashboard 1: University Overview (Landing Page)
 - * Executive KPI summary cards, Top 10 Leaderboard, 5-year score trends, and regional distribution.
- Dashboard 2: Research Analytics
 - * Citations vs Research output scatter plot, Top 10 by Research Productivity Index, citation comparisons.
- Dashboard 3: Student Analytics
 - * International student distribution, faculty-to-student ratio benchmarks, 100% stacked gender diversity.
- Dashboard 4: Country Comparison
 - * National education landscape, country ranking benchmarks, World Bank tertiary indicators vs scores.
- Interlinking Mechanisms:
 - * Standardized Master Keys: university_id (U0001-U2517), country_id (ISO 2-letter), and year.
 - * Dynamic Filter Actions: Selecting University X filters Research and Student dashboards.
 - * Country Propagation: Selecting University X automatically isolates University X's country for benchmarking.

4. Verification, Quality Assurance & Project Delivery

- Data Quality Audit:
 - * 2,517 distinct global universities mapped to standardized country codes.
 - * Missing values treated without blind zero-filling to prevent analytical skew.
 - * Zero duplicate primary keys across dimensional entities.
- Delivery Artifacts:
 - * EduVision_DV.twbx: Production workbook with 13 functional worksheets & 4 integrated dashboards.
 - * eduvision_dashboard_v1.twbx: Phase-1 mid-point development workbook.
 - * eduvision_prototype.twbx: Initial prototype wireframe workbook.
 - * Complete Markdown & PDF technical handbooks, data dictionaries, and cleaning methodologies.
 - * Standardized 5-Milestone GitHub repository structure ready for public portfolio deployment.